

Document Number and Edition: EHIVE03-1918845568-397 ED 003

Document Title: Platform Purification Process Description

SIGNATURES

Function	Name, Position, Department	Signature	Date
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See Responsibilities on next page



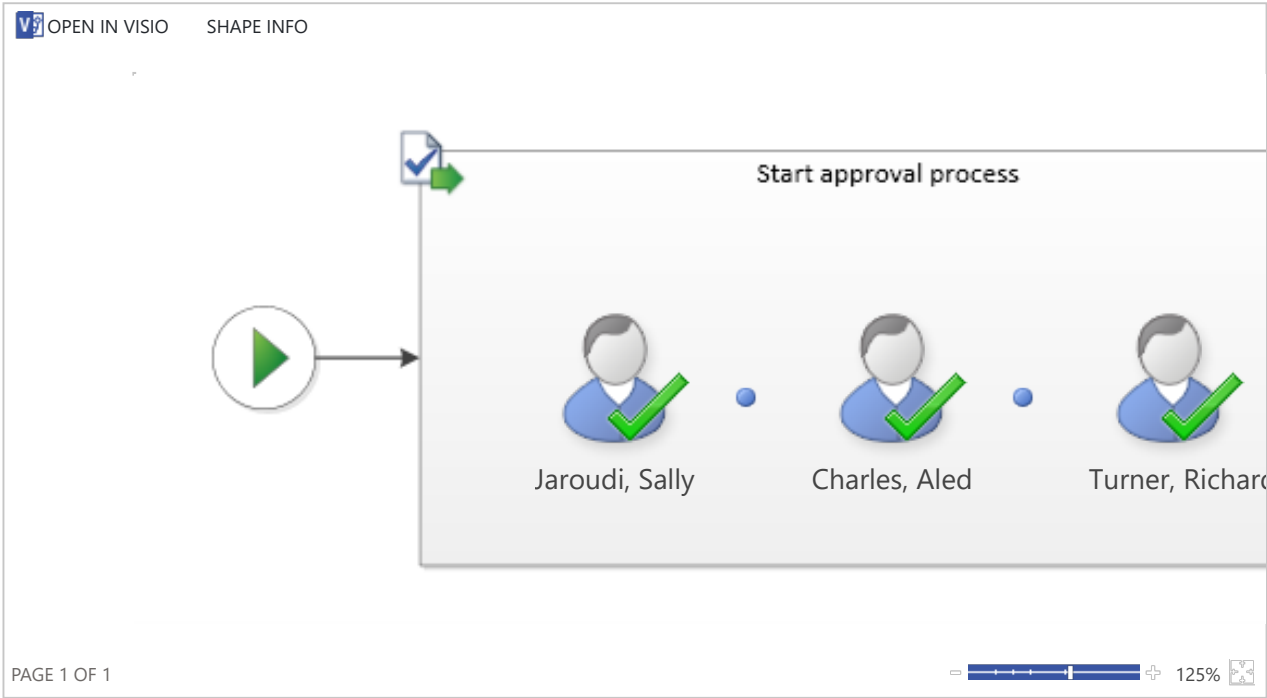
Workflow Status: BPD Reports eApproval

Workflow Information

Initiator: Thilak, Sandhya (Writing Assist Inc.)
Started: 9/26/2019 8:54 AM
Last run: 10/1/2019 5:55 AM

Document: [EHIVE03-1918845568-397 ED 003](#)
Status: Approved

Workflow Visualization



Information about this instance will be automatically removed on 11/30/2019 4:55 AM.

Tasks

This workflow created the following tasks. You can also view them in [Tasks](#).

☐ Assigned To	Title	Due Date	Status	Related Content	Outcome
☐ Jaroudi, Sally	Please approve EHIVE03-1918845568-397 ED 003	9/30/2019	Completed	EHIVE03-1918845568-397 ED 003	Approved
☐ Charles, Aled	Please approve EHIVE03-1918845568-397 ED 003	9/30/2019	Completed	EHIVE03-1918845568-397 ED 003	Approved
☐ Turner, Richard	Please approve EHIVE03-1918845568-397 ED 003	9/30/2019	Completed	EHIVE03-1918845568-397 ED 003	Approved
☐ Pallikal, James	Please approve EHIVE03-1918845568-397 ED 003	9/30/2019	Completed	EHIVE03-1918845568-397 ED 003	Approved

☐ Argoud,
Sarah

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Workflow History

The workflow recorded these events.

☐ Date Occurred	Event Type	User ID	Description	Outcome
9/26/2019 8:54 AM	Workflow Initiated	☐ Thilak, Sandhya (Writing Assist Inc.)	Approval was started. Participants: Jaroudi, Sally;Charles, Aled;Turner, Richard;Pallikal, James;Argoud, Sarah	
9/26/2019 8:54 AM	Task Created	☐ Thilak, Sandhya (Writing Assist Inc.)	Task created for Jaroudi, Sally. Due by: 9/30/2019 12:00:00 AM	
9/26/2019 8:54 AM	Task Created	☐ Thilak, Sandhya (Writing Assist Inc.)	Task created for Charles, Aled. Due by: 9/30/2019 12:00:00 AM	
9/26/2019 8:54 AM	Task Created	☐ Thilak, Sandhya (Writing Assist Inc.)	Task created for Turner, Richard. Due by: 9/30/2019 12:00:00 AM	
9/26/2019 8:54 AM	Task Created	☐ Thilak, Sandhya (Writing Assist Inc.)	Task created for Pallikal, James. Due by: 9/30/2019 12:00:00 AM	
9/26/2019 8:54 AM	Task Created	☐ Thilak, Sandhya (Writing Assist Inc.)	Task created for Argoud, Sarah. Due by: 9/30/2019 12:00:00 AM	
9/26/2019 8:59 AM	Task Completed	☐ Turner, Richard	Task assigned to Turner, Richard was approved by Turner, Richard. Comments:	Approved by Turner, Richard
9/26/2019 10:10 AM	Task Completed	☐ Pallikal, James	Task assigned to Pallikal, James was approved by Pallikal, James. Comments:	Approved by Pallikal, James
9/26/2019 10:25 AM	Task Completed	☐ Charles, Aled	Task assigned to Charles, Aled was approved by Charles, Aled. Comments:	Approved by Charles, Aled
9/26/2019 1:55 PM	Task Completed	☐ Jaroudi, Sally	Task assigned to Jaroudi, Sally was approved by Jaroudi, Sally. Comments:	Approved by Jaroudi, Sally
10/1/2019 5:55 AM	Task Completed	☐ Argoud, Sarah	Task assigned to Argoud, Sarah was approved by Argoud, Sarah. Comments:	Approved by Argoud, Sarah
10/1/2019 5:55 AM	Workflow Completed	☐ Thilak, Sandhya (Writing Assist Inc.)	Approval was completed.	Approval on EHIVE03-1918845568-397 ED 003 has successfully completed. All

participants have completed
their tasks.

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TECHNICAL REPORT
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Function/ Department	Responsibilities
Prepared by: Purification Process Sciences Technology Transfer	Responsible for identifying and/or defining all parameters and pertinent information for transfer to manufacturing as well as the document structure and formatting - Identify all required parameters for transfer to manufacturing based on technology transfer experience to ensure appropriate level of process definition - Maintain consistency across projects and process descriptions - Compile summary tables: BOM, Hold Times, Filtration, Buffers - Verify that targets and ranges for all parameters are defined and are reasonable to maintain process performance within the expected capabilities of facilities
Prepared by: Purification Process Sciences	Responsible for identifying and defining all project specific parameters and values for for transfer to manufacturing - Identify required parameters and define values for targets and ranges for all project specific parameters including buffer recipes based on development experience to ensure appropriate product quality and process performance - Compile process updates and justifications for variations relative to platform process (ED000) or prior process description edition in the Process Changes Section
Approved by: Purification Process Sciences	Responsible for reviewing project specific parameters and confirming that they are in alignment with current PPS strategy (DEV0000_CB_0592) - Verify that project specific parameter targets and ranges are in alignment with general PPS knowledge and expectations - Verify that variations from platform process and prior process are captured in Process Changes and justified
Approved by: Purification Process Sciences Technology Transfer	Responsible for reviewing that all pertinent information/ parameters for transfer to manufacturing are included and in alignment with current BPE strategy - Verify that parameter targets and ranges are in alignment with general PPS knowledge and expectations and facility capabilities - Maintain consistency across projects and process descriptions
Approved by: Manufacturing Sciences	Responsible for reviewing that all pertinent information/ parameters for transfer to manufacturing are included and in alignment with current MS strategy as well as providing any MS specific information contained in the document (i.e. part numbers)

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1 INTRODUCTION

1.1 Purpose

This report describes the platform purification process and process parameters for the production of Drug Substance. Items highlighted in blue are project specific. The overall process (upstream/downstream) is designated “2017 Platform Purification Process”.

Table 1.1-1 General Properties: Drug Substance

Property	Description
Target	TBD
Protein classification	monoclonal antibody (mAb) IgG1
Isoelectric point (pI) range (experimental)	TBD bands between TBD-TBD
Extinction coefficient (experimental)	TBD peaks at pI TBD-TBD (experimentally determined)
Expected titer range (g/L)	TBD (range: TBD-TBD)
Overall expected yield from recovery through Drug Substance (%)	TBD-TBD
Unformulated Drug Substance concentration (g/L)	50
Unformulated Drug Substance formulation	20 mM histidine/histidine-HCl, 240 mM sucrose, pH 6.0
Unformulated Drug Substance presentation	Stored frozen at $\leq -40^{\circ}\text{C}$ in Celsius-Paks
Drug Substance concentration (g/L)	50
Drug Substance formulation	20 mM histidine/histidine-HCl, 240 mM sucrose, 0.02% (w/v) polysorbate 80, pH 6.0
Drug Substance presentation	Stored frozen at $\leq -40^{\circ}\text{C}$ in Celsius-Paks

1.2 Assumptions

- This report does not consider scale or facility constraints.
- Unless otherwise noted, procedures pre/post common unit operations should be followed for filters and unit operations per DEV000_GB_0469.
- Unless otherwise noted, all conductivity values are compensated to a reference temperature of 25°C using a temperature coefficient (θ) of 2.0% per °C.
- The target volumes are expected to be sufficient. Overage to account for system hold up volume is site dependent and should be a separate phase and not added into the volume targets for the process.
- Unless otherwise noted, flow rate ranges are for steady state operation and do not consider pump ramping strategy.
- Unless otherwise noted, all unit operations are conducted at controlled room temperature (15-27°C).
- For facilities for which 1.0 N NaOH is not feasible, 0.5 N NaOH may be used as a substitute with “Conductivity (mS/cm @ 25°C), final” NOR: 84 – 113.
- Unless otherwise noted, all chromatography step phases are operated in down flow mode.
- Unless otherwise noted, all process parameters are inputs.

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1.3 Abbreviations/Acronyms/Definition

Table 1.3-1 Abbreviations and Acronyms

Acronym	Definition	Acronym	Definition
AEX	Anion exchange chromatography	LG	Linear gradient
AL	Alert limit	LR	Lot release
Bldg	Building	LMH	Liters/meter ² -hour
BR	Batch record	MV	Membrane volume
CEX	Cation exchange chromatography	MWCO	Molecular Weight Cut Off
CM	Conditioned media	NA	Not applicable
Conc.	Concentration	NC	Non-conformance
CV	Column volume	NF	National Formulary
DV	Diavolumes	NOR	Normal operating range
DBC	Dynamic binding capacity	OD	Optical density
DCF	Density Correction Factor	PCR	Polymerase chain reaction
DS	Drug Substance	PES	Polyethersulfone
DSC	Development Scale-Up Center (500-L scale)	PF	Product Filter
EAB	Excipient addition buffer	PFI	Pre-formulated intermediate
ELISA	Enzyme-linked immunosorbent assay	P/N	Part number
EP	European Pharmacopeia	PP	Polypropylene
FB	Formulated bulk	psid; psig	pounds per square inch differential; pounds per square inch gauge
FIO	For information only	PS80	Polysorbate 80
FF	Fast Flow	Protein A	Protein A affinity chromatography
FSS	Formulation Spiking Solution	qs	Bring to final weight (quantum satis)
FMC	Frederick Manufacturing Center	RPHPLC	Reverse phase high performance chromatography
GPF	Gaithersburg Pilot Facility	S.F.	Safety factor
GMP	Good Manufacturing Practices	SS	Stainless steel
HETP	Height equivalent to a theoretical plate	SUL	Scale-Up Lab
HIC	Hydrophobic interaction chromatography	TBD	To be determined
HPLC	High-performance liquid chromatography	TMP	Trans-membrane pressure
HPSEC	High-performance size-exclusion chromatography	UB	Unformulated Bulk
HCP	Host cell protein	UDS	Unformulated Drug Substance
IgG1/2/4	Immunoglobulin G1/2/4	USP	United States Pharmacopoeia
IEC	Ion exchange chromatography	VC	Viral clearance
JP	Japanese Pharmacopoeia	VF	Virus filtration

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Table 1.3-1 Abbreviations and Acronyms

Acronym	Definition	Acronym	Definition
kDa	Kilodalton	VPro	Viresolve Pro
LF	Load Filter	WFI	Water for injection

1.4 Definitions of Terminologies for Parameter Ranges

1.4.1 Target

The target value is a preset value that a process parameter is aimed to by process controls. The controls include any combination of manual or automated, discrete or continuous, feed-forward or feed-back responses to operate the process near the target value. Where Targets are provided, the process should be controlled to meet the target value as closely as possible, not just controlling to the edge of the Alert Limit or Normal Operating Range.

1.4.2 Alert Limit (AL)

This is the expected process range. Exceeding this limit suggests that corrective action is required to prevent the process from exceeding the normal operating range (NOR) or other process constraints. Development and subject matter experts should be informed to determine if any corrective action is required. Exceeding this limit does not trigger a non-conformance investigation.

1.4.3 Normal Operating Range (NOR)

Operating parameter ranges, also called Action Limits, which are established based on current understanding and process knowledge gained through development and process characterization. NORs are specified in manufacturing batch records and/or other GMP documentation (such as automation recipes) to control the process. A value exceeding a NOR triggers a non-conformance investigation and is documented and evaluated according to the site's Quality System.

2 BILL OF MATERIALS

Table 2-1 Bill of Materials

Type	Component Name	Minimum Grade	MEDI P/N	Justification/Notes
Chromatography media	MabSelect SuRe (Protein A)	NA	4101797 4102160 6001979 6002000	5 L (ethanol) 10 L (ethanol) 5 L (benzyl) 10 L (benzyl)
	Capto Q (AEX)	NA	6001609 6001621	NA
	Poros 50 HS (CEX)	NA	40003 4102211	NA
Buffer and solution components	Acetic acid, glacial	USP, EP	4101341 6000325	NA
	Benzyl alcohol	NF, EP, JP	4102670	NA
	L-histidine	USP, EP, JP	6001825	NA
	L-histidine HCl, monohydrate	EP, JP	4101011	NA
	Polysorbate 80	NF, EP	4100792	NA
	Sodium acetate, trihydrate, crystal	USP, EP, JP	4101342 6000326	NA
	Sodium chloride	USP, EP	6000324	NA
	10N Sodium hydroxide	NA	4102099 4102229	Biotech Reagent
	Sodium hydroxide, pellets	EP, NF	4101097	NA
	Sucrose (low endotoxin)	EP, NF, JP	4102624	NA
	Tris base (tromethamine/trometamol)	USP, EP	4101399	NA
	Tris hydrochloride	NA	4100924	Biotech Reagent
Purification filters	Viresolve Pro Magnus Device	NA	4102501 4102503	0.51 m ² 1.53 m ²
	Viresolve Pro Magnus Shield	NA	4102530 4102531	0.51 m ² 1.53 m ²
	Pellicon 3 Ultracel C Screen ultrafiltration cassettes, 30kDa MWCO	NA	6001576 4102051 4102050	0.11 m ² 0.57 m ² 1.14 m ²
	Millistak+ HC depth Pod filters-C0HC	NA	4101552	1.1 m ²
	Durapore Opticap, 0.22 µm	NA	NA	Sterile Grade 0.2 µm
	Millipore Express SHC, 0.5/0.2 µm	NA	NA	Generic 0.2 µm
	Pall Fluorodyne EX EDF, 0.2 µm	NA	NA	Generic 0.2 µm
Miscellaneous components	Celsius-Pak 16.6 L with KPC Connectors	NA	4101227	NA

3 IN-PROCESS HOLD TIMES

Process intermediate stability studies were performed to support pooling of chromatography cycles and holding of process intermediates.

Hold times begin when product is collected and end when the next step is loaded.

Unless otherwise noted, room temperature hold times are inclusive to refrigerated temperature hold times but not vice versa. For example, a product that was held for 48 hours at room temperature prior to being stored 24 hours refrigerated would have a total hold time of 72 hours at refrigerated temperature but only 48 hours at room temperature.

Table 3-1 Process Intermediate Hold Times

Process Intermediate	Room Temperature Hold Time - Max (days, 15-25 °C)	Refrigerated Temperature Hold Time - Max (days, 2-8 °C)	Justification/Notes
Conditioned medium	14	28	NA
Protein A Product	14	28	NA
Virus Inactivation Product	14	28	NA
AEX Product	14	28	NA
CEX Load Adjustment Product	14	28	NA
CEX Product	14	28	NA
Virus Filtration Product	14	28	Leverages CEX Product Data
Pre-formulated Bulk	14	28	NA
Formulated Bulk	14	28	NA

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4 LOAD AND PRODUCT FILTRATION

Table 4-1 Filtration Parameters

Unit Operation	Position	Membrane Type	In-line	Volumetric Load NOR-Max (L/m ²)	Justification/Notes
Protein A Chromatography (LF → Unit Op → PF)	Load Filter	Express SHC, 0.5/0.2 µm	All Steps	3000	NA
	Product Filter	Express SHC, 0.5/0.2 µm	Collection	3000	NA
Virus Inactivation (Unit Op → PF 1 → PF 2)	Product Filter 1	Millistak+ HC POD filters - C0HC	Off-line	500	NA
	Product Filter 2	Express SHC, 0.5/0.2 µm	Off-line	3000	NA
AEX (LF → Unit Op → PF)	Load Filter	Express SHC, 0.5/0.2 µm	All Steps	3000	NA
	Product Filter	Express SHC, 0.5/0.2 µm	Collection	3000	NA
CEX (LF → Unit Op → PF)	Load Filter	Express SHC, 0.5/0.2 µm	All Steps	3000	NA
	Product Filter	Express SHC, 0.5/0.2 µm	Collection	3000	NA
Virus Filtration (Unit Op → PF)	Product Filter	Express SHC, 0.5/0.2 µm	Collection	3000	NA
Concentration and Diafiltration (Unit Op → PF)	Product Filter	Express SHC, 0.5/0.2 µm	Off-line	500	Recommend smallest device within range
Drug Substance Formulation and Bulk Filtration (Unit Op → Bulk Filter)	Bulk Filtration	Durapore Opticap, 0.2 µm	Off-line	500	Recommend smallest device within range

Millipore SHC filters may be interchanges with Pall Fluorodyne EX EDF filters of the same device size. This assumes the additional filter area of the Pall device compensates for any difference in maximum volumetric load.

5 PROTEIN A CHROMATOGRAPHY

Table 5-1 Protein A Chromatography: Process Parameters

Step	Process Parameter (units)	Target	AL	NOR/Action Limit	Justification/Notes
General	Media	MabSelect SuRe			Prep per appropriate SOP
Equilibration 1 <i>50 mM tris, pH 7.4</i>	Linear velocity (cm/hr)	300	285 – 315	≤ 330	AL is ± 5% of set-point
	Volume (CV)	3.0	2.9 – 4.0	≥ 2.5	NA
	pH, effluent, (endpoint check) ^a	NA	7.2 – 7.6	7.0 – 7.8	NA
	Conductivity, effluent, (endpoint check) (mS/cm @ 25°C) ^a	NA	NA	NA	Run and record.
Load <i>Conditioned Medium</i>	Linear velocity (cm/hr)	300	285 – 315	≤ 330	AL is ± 5% of set-point
	Protein load (g/L resin)	NA	20 – 40	≤ 44	NOR-Max is the capacity. AL-Min is based on productivity. AL-Max is used to calculate number of cycles.
Equilibration 2 <i>50 mM tris, pH 7.4</i>	Linear velocity (cm/hr)	300	285 – 315	≤ 330	AL is ± 5% of set-point
	Volume (CV)	3.0	2.9 – 4.0	≥ 2.5	Add depth filter hold-up volume to the target
Wash <i>50 mM tris, 0.5 M sodium chloride, pH 7.4</i>	Linear velocity (cm/hr)	300	285 – 315	≤ 330	AL is ± 5% of set-point
	Volume (CV)	3.0	2.9 – 4.0	≥ 2.5	NA
Equilibration 3 <i>50 mM tris, pH 7.4</i>	Linear velocity (cm/hr)	300	285 – 315	≤ 330	AL is ± 5% of set-point
	Volume (CV)	3.0	2.9 – 4.0	≥ 2.5	NA
Elution <i>25 mM sodium acetate, pH 3.6</i>	Linear velocity (cm/hr)	150	142 – 158	≤ 330	AL is ± 5% of set-point
	Product collection start (OD ₂₈₀)	0.5	0.4 – 1.0	≥ 0.3	Volume before peak = ~ 1.0 CV

Table 5-1 Protein A Chromatography: Process Parameters

Step	Process Parameter (units)	Target	AL	NOR/Action Limit	Justification/Notes
	Product collection end (OD ₂₈₀)	0.5	0.4 – 1.0	≥ 0.3	Product volume = ~ 2.0 CV Step yield = ~ 95%
Acid Strip <i>100 mM acetic acid</i>	Linear velocity (cm/hr)	300	285 – 315	≤ 330	AL is ± 5% of set-point
	Volume (CV)	3.0	2.9 – 4.0	≥ 2.5	NA
Equilibration 4 <i>50 mM tris, pH 7.4 or WFI</i>	Linear velocity (cm/hr)	300	285 – 315	≤ 330	AL is ± 5% of set-point
	Volume (CV)	1.0	0.9 – 2.0	≥ 0.5	NA
NaOH Strip <i>0.1 N sodium hydroxide</i>	Linear velocity (cm/hr)	300	285 – 315	≤ 330	AL is ± 5% of set-point
	Volume (CV)	3.0	2.9 – 4.0	≥ 2.5	Sanitization buffer
	Conductivity, effluent, (endpoint check) (mS/cm @ 25°C) ^a	NA	NA	19 – 27	NA
End of Cycle					
Sanitization Hold	Contact time (minutes)	35	30 – 60	30 – 120	NA
Storage <i>2% (v/v) benzyl alcohol, 100 mM sodium acetate, pH 5.0</i>	Linear velocity (cm/hr)	300	285 – 315	≤ 330	AL is ± 5% of set-point
	Volume (CV)	3.0	2.9 – 4.0	≥ 2.5	NA
	pH, effluent, (endpoint check) ^a	NA	NA	4.8 – 5.2	NA

^a Process output

6 VIRUS INACTIVATION (LOW PH TREATMENT)

Table 6-1 Virus Inactivation: Process Parameters

Step	Process Parameter (units)	Target	AL	NOR/Action Limit	Justification/Notes
Acid Titration <i>1 M acetic acid</i>	pH, (endpoint check) ^a	3.5	NA	3.4 – 3.6	~ 0.10 (kg acid / kg Product)
Incubation	Duration (minutes)	30	15 – 60	15 – 240	NOR-Min is based on viral clearance recommendation. NOR-Max is based on development stability data.
	Temperature (°C)	NA	NA	15 – 25	Confirm temperature at start and end of incubation
	pH	NA	NA	3.4 – 3.6	Confirm pH at start and end of incubation
Base Titration <i>1 M tris base</i>	pH, (endpoint check) ^a	7.4	7.3 – 7.5	7.2 – 7.6	(kg base / kg acid) ~ 1.0 Step yield = ~ 100%

^a Process output

7 AEX

Table 7-1 AEX: Process Parameters

Step	Process Parameter (units)	Target	AL	NOR/Action Limit	Justification/Notes
General	Media	Capto Q			Prep per appropriate SOP
Pre-Equilibration <i>50 mM tris, 0.5 M sodium chloride, pH 7.4</i>	Linear velocity (cm/hr)	300	285 – 315	≤ 330	AL is ± 5% of set-point
	Volume (CV)	5.0	4.9 – 6.0	≥ 4.5	NA
Equilibration <i>50 mM tris, pH 7.4</i>	Linear velocity (cm/hr)	300	285 – 315	≤ 330	AL is ± 5% of set-point
	Volume (CV)	3.0	2.9 – 4.0	≥ 2.5	NA
	pH, effluent, (endpoint check) ^a	NA	NA	7.2 – 7.6	NA
	Conductivity, effluent, (endpoint check) (mS/cm @ 25°C) ^a	NA	NA	NA	Run and record.
Load <i>Virus Inactivation Product</i>	Linear velocity (cm/hr)	300	285 – 315	≤ 330	AL is ± 5% of set-point
	Protein load (g / L resin)	NA	50 – 100	≤ 110	NOR-Max is based on VC study. AL-Min is based on productivity. AL-Max is used to calculate number of cycles.
	Product collection start (OD ₂₈₀)	0.5	0.4 – 1.0	≥ 0.3	Volume before peak = ~ 1.0 CV
Chase <i>50 mM tris, pH 7.4</i>	Linear velocity (cm/hr)	300	285 – 315	≤ 330	AL is ± 5% of set-point
	Product collection emd (OD ₂₈₀)	0.5	0.4 – 1.0	≥ 0.3	Product dilution = ~ 10% Step yield = ~ 95%

Table 7-1 AEX: Process Parameters

Step	Process Parameter (units)	Target	AL	NOR/Action Limit	Justification/Notes
Salt Strip <i>50 mM tris, 0.5 M sodium chloride, pH 7.4</i>	Linear velocity (cm/hr)	300	285 – 315	≤ 330	AL is $\pm 5\%$ of set-point
	Volume (CV)	3.0	2.9 – 4.0	≥ 2.5	NA
NaOH Strip <i>1.0 N sodium hydroxide</i>	Linear velocity (cm/hr)	300	285 – 315	≤ 330	AL is $\pm 5\%$ of set-point
	Volume (CV)	3.0	2.9 – 4.0	≤ 2.5	Sanitization buffer
	Conductivity, effluent, (endpoint check) (mS/cm @ 25°C) ^a	NA	NA	164 – 211	NA
End of Cycle					
Sanitization Hold	Contact time (minutes)	35	30 – 60	15 – 120	NA
Storage <i>0.1 N sodium hydroxide</i>	Linear velocity (cm/hr)	300	285 – 315	≤ 330	AL is $\pm 5\%$ of set-point
	Volume (CV)	3.0	2.9 – 4.0	≥ 2.5	NA
	Conductivity, effluent, (endpoint check) (mS/cm @ 25°C) ^a	NA	NA	19 – 27	NA

^a Process output

8 AEX PRODUCT ADJUSTMENT

Table 8-1 AEX Product Adjustment: Process Parameters

Step	Process Parameter (units)	Target	AL	NOR/Action Limit	Justification/Notes
pH Adjustment <i>1 M acetic acid</i>	pH	5.0	4.9 – 5.1	4.8 – 5.2	~ 0.05 (kg titrant / kg AEX Product)

9 CEX

Table 9-1 CEX: Process Parameters

Step	Process Parameter (units)	Target	AL	NOR/Action Limit	Justification/Notes
General	Media	Poros HS 50			Prep per appropriate SOP
Equilibration 1 <i>50 mM sodium acetate, pH 5.0</i>	Linear velocity (cm/hr)	300	285 – 315	≤ 330	AL is ± 5% of set-point
	Volume (CV)	3.0	2.9 – 4	≥ 2.5	NA
	pH, effluent, (endpoint check) ^a	NA	NA	4.8 – 5.2	NA
	Conductivity, effluent, (endpoint check) (mS/cm @ 25°C) ^a	NA	NA	NA	Run and record.
Load <i>pH Adjusted AEX Product</i>	Linear velocity (cm/hr)	300	285 – 315	≤ 330	AL is ± 5% of set-point
	Protein load (g / L resin)	NA	25 – 50	≤ 55	NOR-Max is the capacity. AL-Min is based on productivity. AL-Max is used to calculate number of cycles.
Equilibration 2 <i>50 mM sodium acetate, pH 5.0</i>	Linear velocity (cm/hr)	300	285 – 315	≤ 330	AL is ± 5% of set-point
	Volume (CV)	3.0	2.9 – 4.0	≥ 2.5	NA
Elution <i>50 mM sodium acetate, 200 mM sodium chloride pH 5.0</i>	Linear velocity (cm/hr)	150	142 – 158	≤ 330	AL is ± 5% of set-point
	Product collection start (OD ₂₈₀)	0.5	0.4 – 1.0	≥ 0.3	Volume before peak = ~ 1.0 CV
	Product collection end (OD ₂₈₀)	0.5	0.4 – 1.0	≥ 0.3	Product volume = ~ 3.0 CV Step yield = ~ 90%

Table 9-1 CEX: Process Parameters

Step	Process Parameter (units)	Target	AL	NOR/Action Limit	Justification/Notes
NaOH Strip <i>1.0 N sodium hydroxide</i>	Linear velocity (cm/hr)	300	285 – 315	≤ 330	AL is ± 5% of set-point
	Volume (CV)	3.0	2.9 – 4	≤ 2.5	Sanitization buffer
	Conductivity, effluent, (endpoint check) (mS/cm @ 25°C) ^a	NA	NA	164 – 211	NA
End of Cycle					
Sanitization Hold	Contact time (minutes)	35	30 – 60	15 – 120	NA
Storage <i>0.1 N sodium hydroxide</i>	Linear velocity (cm/hr)	300	285 – 315	≤ 330	AL is ± 5% of set-point
	Volume (CV)	3.0	2.9 – 4.0	≤ 2.5	NA
	Conductivity, effluent, (endpoint check) (mS/cm @ 25°C) ^a	NA	NA	19 – 27	NA

^a Process output

10 VIRUS FILTRATION

Table 10-1 Virus Filtration: Process Parameters

Step	Process Parameter (units)	Target	AL	NOR/Action Limit	Justification/Notes
General	Media		Viresolve Pro +		Prep per appropriate SOP
	Feed pressure (psig)	NA	0 – 55	0 – 60	NA
	Differential pressure (psid)	30	0 – 35	0 – 60	
	Volumetric load (L/m ²)	75	≥ 70	≥ 65	NA
Equilibration 50 mM sodium acetate, 200 mM sodium chloride pH 5.0	pH, effluent, (endpoint check) ^a	NA	4.8 – 5.2	NA	Run and record.
	Differential pressure (psid)	30	0 – 35	0 – 45	Based on process Acceptable Range maximum of 50 psid per report: <i>Justification of Upper Pressure Limit for VPro Virus Filter Operation</i>
	Total volumetric load (L/m ² filter area)	NA	≤ 1000	≤ 1100	NOR-Max is based on VC study. AL-Max is used to calculate filter area. Measured from the start of product load to the end of post-filtration flush (Equilibration 2)
	Differential pressure (psid)	30	0 – 35	0 – 45	
Load CEX Chromatography Product	Flow decay (%) ^a	NA	≤ 75	≤ 90	100% × (Initial buffer flux – Final virus filtration flux) ÷ (Initial buffer flux)
	Product collection start (OD ₂₈₀)	0.5	0.4 – 1.0	NA	NA
	Differential pressure (psid)	30	0 – 35	0 – 45	Based on process Acceptable Range maximum of 50 psid per report: <i>Justification of Upper Pressure Limit for VPro Virus Filter Operation</i>
	Product collection end (OD ₂₈₀)	0.5	0.4 – 1.0	NA	Volume chase may be set if OD is not available. Step yield = ~ 100%

^a Process output.

11 CONCENTRATION AND DIAFILTRATION

Table 11-1 Concentration and Diafiltration: Process Parameters

Step	Process Parameter (units)	Target	AL	NOR/Action Limit	Justification/Notes
General	Media	Pellicon 3 Ultracel C Screen 30kDa MWCO			Prep per appropriate SOP
	Feed pressure (psig)	NA	≤ 55	≤ 60	NA
Equilibration <i>20 mM histidine/histidine-HCl, pH 5.9</i>	pH, permeate (endpoint check) ^a	NA	5.8 – 6	5.7 – 6.1	NA
Concentration 1 Control Method: Constant TMP and Feed Flow Control	Membrane protein load (g/m ²)	NA	500 – 1000	≤ 1100	(Mass of protein)/(area of ultrafiltration membrane) NOR-Max is the tested capacity. AL-Min is based on productivity. AL-Max is used to calculate filter area.
	TMP (psi)	20	≤ 30	≤ 35	NA
	Fed batch ratio	NA	NA	NA	(Load volume)/(DCF tank volume)
	Feed flow (L/min/m ²)	5.0	4.0 – 6.0	3.0 – 7.0	May lower to maintain constant TMP.
	Protein concentration, endpoint check (g/L)	70.0	60 – 80	≤ 110	May be directly measured or calculated from permeate or retentate volume
Diafiltration <i>20 mM histidine/histidine-HCl, pH 5.9</i> Control Method: Constant TMP and Feed Flow Control	TMP (psid)	20	≤ 30	≤ 35	NA
	Feed flow (L/min/m ²)	5.0	4.0 – 6.0	3.0 – 7.0	May lower to maintain constant TMP
	Number of diavolumes (DV)	6.0	5.5 – 8.0	≥ 5.5	(kg buffer)/(kg product)
	pH, permeate, (endpoint check) ^a	NA	5.8 – 6.0	NA	NA
	pH, retentate, (endpoint check) ^a	NA	NA	5.8 – 6.2	NA

Table 11-1 Concentration and Diafiltration: Process Parameters

Step	Process Parameter (units)	Target	AL	NOR/Action Limit	Justification/Notes
Concentration 2 Control Method: Constant TMP and Feed Flow Control	TMP (psid)	20	≤ 30	≤ 35	NA
	Feed flow (L/min/m ²)	5.0	≤ 6.0	≤ 7.0	May lower to maintain constant TMP
	Protein concentration, endpoint check (g/L)	100	90 – 110	63.4 - 110	May be directly measured or calculated from permeate or retentate volume
Recirculation	Feed pressure (psig)	25	≤ 55	≤ 60	NA
	Feed flow (L/min/m ²)	NA	≤ 6	≤ 7	NA
	Recirculation time (min)	5	≤ 30	≤ 120	NA
Recovery Flush <i>20 mM histidine/histidine-HCl, pH 5.9</i> Recovery Method: Buffer flush of system hold up volume	Feed pressure (psi)	15	≤ 55	≤ 60	NA
	Feed flow (L/min/m ²)	NA	≤ 6.5	≤ 7	NA
	Protein concentration, endpoint check (g/L)	85.0	76.5 – 93.5	63.4 – 110	NA
Dilution to Pre-formulated Intermediate <i>20 mM histidine/histidine-HCl, pH 5.9</i>	Protein concentration, PFI (g/L)	66.7	65 – 68.4	63.4 – 70.0	NA
	pH, (endpoint check) ^a	NA	5.9 – 6.1	5.8 – 6.2	NA

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Table 11-2 Concentration and Diafiltration: Density Values

Buffer	Step	Protein Concentration (mg/mL)	Density (g/mL)	Notes
50 mM sodium acetate, 200 mM sodium chloride, pH 5.0	Conc. 1	0 – 10	1.011	Density = (0.000285 × protein concentration) + buffer density
	Conc. 1	10 – 20	1.014	NA
	Conc. 1	20 – 30	1.017	NA
	Conc. 1	30 – 40	1.020	NA
	Conc. 1	40 – 50	1.023	NA
	Conc. 1	50 – 60	1.026	NA
	Conc. 1	60 – 70	1.029	NA
20 mM histidine/histidine-HCl, pH 5.9	PFI	66.7	1.020	NA
	Diafiltration	70	1.021	NA
	Conc. 2	70 – 80	1.022	NA
	Conc. 2 (Recovery)	80 – 90 (85)	1.025	NA
	Conc. 2	90 – 100	1.028	NA
	Conc. 2, final	100	1.030	NA

12 FORMULATION

Table 12-1 Formulation: Process Parameters

Step	Process Parameter (units)	Target	AL	NOR/Action Limit	Justification/Notes
EAB Spike (of PFI to UB) <i>EAB:</i> 20 mM histidine/histidine-HCl, 960 mM sucrose, pH 6.0 (density 1.127 g/mL)	Ratio of EAB spike to PFI (kg/kg)	0.373	0.369 – 0.377	0.364 – 0.382	UB/UDS: 50 g/L MEDXXXXX , 20 mM histidine/histidine-HCl, 240 mM sucrose, pH 6.0 (theoretical density 1.039 g/mL) Proceed to Bulk Filtration for UDS
	Protein concentration, endpoint check (g/L)	50.0	47.5 – 52.5	N/A	NA
	Osmolality (mOsm/kg) ^a	300	255 – 345	NA	NA
FSS Spike (of UB to FB) <i>FSS:</i> 10% (w/v) polysorbate 80 (density 1.01 g/mL)	Ratio of FSS spike to UB (g/kg)	1.928	1.909 – 1.947	1.87 – 1.986	FB/DS: 50 g/L MEDXXXXX , 20 mM histidine/histidine-HCl, 240 mM sucrose, 0.02% (w/v) polysorbate 80, pH 6.0 (theoretical density 1.039 g/mL) Optional, perform prior to Bulk Filtration for DS

13 BULK FILTRATION

Table 13-1 Bulk Filtration: Process Parameters

Step	Process Parameter (units)	Target	AL	NOR/Action Limit	Justification/Notes
0.2 µm Bulk Filtration	Drug Substance container	Celsius-Paks			NA
	Fill volume per container (L)	NA	NA	5.15 – 16.6	If multiple containers are required, target equivalent volume in each.
	Drug Substance storage temperature (°C)	5	NA	2 – 8	Drug Substance can be held 6 months at 2-8°C prior to shipment.
Integrity Test 0.2 µm Filter	Integrity Test Result (PASS/FAIL) ^a	PASS	NA	PASS	Perform per appropriate SOP. Integrity test is an in-process control that must pass for the lot to be released

14 BUFFERS AND SOLUTIONS

Buffer and solution stability studies were performed by holding solutions and buffers at ambient temperature in stainless steel containers. The studies were designed to test chemical stability for longer than the expected maximum time for routine storage. These times may be extended by an appropriate study unless otherwise noted. The studies do not address microbial control, which should be defined by the facility. Assays used to measure stability included pH and conductivity. In some cases, the hold time is measured directly (direct). In other cases, the hold time is supported by the hold time of similar buffers if the proposed buffer falls within our current buffer matrix (matrix). Please see DEV000_GB_0533 for reference.

Table 14-1 Buffer and Solution Stability

Buffer Name	Recipe Table	MEDI PN/ Doc Num	Hold Time - Max (days, 15-25°C)	Unit Operation	Steps
50 mM tris, pH 7.4	DEV000_GB_0469 10-1	8003734/ A-005378	35 (direct)	Protein A	Equilibration 1
					Equilibration 2
					Equilibration 3
					Equilibration 4
				Virus Inactivation	Depth Filter Equilibration
50 mM tris, 0.5 M sodium chloride, pH 7.4	DEV000_GB_0469 10-2	8003737/ A-005380	35 (matrix)	AEX	Depth Filter Chase
					Equilibration 1
					Chase
				Protein A	Wash
25 mM sodium acetate, pH 3.6	DEV000_GB_0469 10-3	8002814/ A-005286	30 (matrix)	AEX	Pre-Equilibration
				AEX	Salt Strip
				CEX	Salt Strip
100 mM acetic acid	DEV000_GB_0469 10-4	A-005041/ A-005041	30 (direct)	Protein A	Elution
					Acid Strip

Table 14-1 Buffer and Solution Stability

Buffer Name	Recipe Table	MEDI PN/ Doc Num	Hold Time - Max (days, 15-25°C)	Unit Operation	Steps
WFI	NA	NA	NA	Protein A	Equilibration 4
0.1 N sodium hydroxide	DEV000_GB_0469 10-5	A0014-D/ A0014-D	TBD	Protein A	NaOH Strip
				AEX	Storage
				CEX	Storage
2% (v/v) benzyl alcohol, 100 mM sodium acetate, pH 5.0	DEV000_GB_0469 10-6	8002256/ A-005212	30 (direct)	Protein A	Storage
1 M acetic acid	DEV000_GB_0469 10-7	A-005168/ A-005168	77 (direct)	Virus Inactivation	Acid Titration
1 M tris base	DEV000_GB_0469 10-8	A0101-D/ A0101-D	35 (matrix)	AEX Product Adjustment	Adjustment
				Virus Inactivation	Base Titration
50 mM sodium acetate, pH 5.0	DEV000_GB_0469 10-9	A-005097/ A-005097	35 (direct)	CEX	Equilibration 1
50 mM sodium acetate, 200 mM sodium chloride, pH 5.0	14-2	8003774/ A-005389	TBD	CEX	Equilibration 2
					Elution
					Equilibration 1
1.0 N sodium hydroxide	DEV000_GB_0469 10-10	A0191-D/ A0191-D	180 (direct)	AEX	NaOH Strip
20 mM histidine/histidine-HCl, pH 5.9	DEV000_GB_0469 10-19	8005030/ A-005478	TBD	Concentration and Diafiltration	NaOH Strip
					Equilibration
					Diafiltration
20 mM histidine/histidine-HCl, 960 mM sucrose, pH 6.0	DEV000_GB_0469 10-20	8004947/ A-005465	TBD	Formulation	Recovery Flush
				Dilution to PFI	
				EAB Spike	

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Table 14-1 Buffer and Solution Stability

Buffer Name	Recipe Table	MEDI PN/ Doc Num	Hold Time - Max (days, 15-25°C)	Unit Operation	Steps
10% (w/v) polysorbate 80	DEV000_GB_0469 10-11	A-0647/ A-0647	10 ^a	Formulation	FSS Spike

^a DEV000_GB_0303 ED 001; Document Title: Stability Study Report for 10% Polysorbate 80 Aqueous Stock Solution: The results demonstrate that the 10% PS80 (w/v) stock solution can be stored in the dark for 30 days at 2-8°C and 10 days at 23-27°C. At these storage conditions, the PS80 stock solution meets United States Pharmacopoeias' (USP), European (Ph. Eur.) and Japanese (JP) specifications.

Table 14-2 50 mM sodium acetate, 200 mM sodium chloride, pH 5.0

MEDI PN/ Document No.	8003774/A-005389		
Density Corrected	Yes		
Specification	Design Target	NOR/Action Limit	
pH	5.0	4.9 – 5.1	
Conductivity (mS/cm at 25°C)	NA	NA	
Osmolality (mOsm/kg)	NA	NA	
Density (g/mL)	1.01	NA	
Chemical	MEDI PN	Amount by Volume	Amount by Weight
Sodium acetate, trihydrate	4101342 6000326	NA	4.87 g/kg
Acetic acid, glacial	4101341 6000325	0.78 mL/kg	0.82 g/kg
Sodium chloride	6000324	NA	11.57 g/kg
WFI	NA	NA	0.983 kg/kg
Total	1000 g/kg		

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15 PROCESS CHANGES

Table 15-1 lists changes in the Platform Purification Process Description from edition 001 through edition 003. These changes do not warrant a process number change from 2017.

Table 15-1 2017 to TBD Change Summaries and Justifications

Edition	Unit Operation	Step	Parameter	Original	New	Summary/ Justification
002	Protein A	Strip	NA	NA	Added	Acid strip implemented to ensure proper cleaning of column prior to NaOH Strip
003	Virus Filtration	Load, Chase	Differential Pressure	NOR max = 50 psid	NOR max = 45 psid	Process Acceptable Range maximum = 50 psid per report. <i>Justification of Upper Pressure Limit for VPro Virus Filter Operation</i>

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16 EDITION HISTORY

Edition	Reason
001	New document
002	Update to current template See Section 15 Process Changes
003	Updated nomenclature throughout document to align with Cycle 3 process development nomenclature per EHIVE03-1918845568-649 Merged AL min and max columns into one column – ease document’s transition from cycle 1 to cycle 3 template Merged NOR min and max columns into one column – ease document’s transition from cycle 1 to cycle 3 template See section 15 Process Changes